

Data Structure and Algorithms

- Data Structure and algorithms are very crucial in programming because they help us work with data. A data structure is a way of organizing data, while an algorithm is a set of steps used to solve a problem or do a task. For example, we can use a data structure to store a list of names and an algorithm to find a specific name from that list

Linear Data Structure

- A Linear Data Structure is a way of storing data where the items are placed one after another in a line or order. Each item has a place in the sequence. Some examples are arrays, linked lists, stack, and queues. It is called linear because we can follow the data from one item to the next. This type of structure is useful when we want to keep the data organized.

One-Dimensional arrays

- A One-dimensional array is a group of data stored in one line or sequence. Each time has a number called an index. Which tells us its position. The index starts at 0. For example if we have a [5,55,555] the 5 was 0 and the 55 was 1 while the 555 was 2. Arrays are useful for keeping related information together and making it easier to find a specific item

The big O Notation

- Big O notation is used to show how fast or slow an algorithm can be when the amount of data gets bigger. It helps us understand how much work an algorithm needs to do. For example $O(n)$ means that the work can increase as the number of data increases. Big O is useful when comparing algorithms it helps us know which one can handle more data.

Basic Searching Algorithms

- Searching algorithms are implemented when we have to find out an element among many other elements. For instance, if there is a set of elements and you need to find out element no. 50 from this set of elements, then a searching algorithm can be utilized in such a scenario. There are two types of search algorithms, linear and binary. In a linear search algorithm, elements are checked one after another, while in binary search, the middle element is checked, and half of the options are discarded at once.

Linear Search

- In linear search, an item is searched one by one from the beginning to find the required item. The process continues until either the item is found or the whole set of items is checked. For instance, if we want to search 30 in [10, 20, 30, 40], we will first search 10, then 20, then 30. Linear search can be performed on both unsorted and sorted data, although the process may take some time in the case of large amounts of data.

Binary Search

- Binary search algorithm is applied to identify a certain item within a sorted set of items. Rather than going through all the items in the list, binary search algorithm will begin with the middle item. When the sought item is not found, we determine whether to continue the search on either the left side or the right side of the item

REFERECES

GeeksforGeeks. (n.d.). *Big O analysis*.

<https://www.geeksforgeeks.org/dsa/analysis-algorithms-big-o-analysis/>

GeeksforGeeks. (n.d.). *Introduction to linear data structures*.

<https://www.geeksforgeeks.org/dsa/introduction-to-linear-data-structures/>

GeeksforGeeks. (2026, March 3). *Searching algorithms*.

<https://www.geeksforgeeks.org/dsa/searching-algorithms/>

Ghosh, A. K. (2025, September 23). *One dimensional array in data structures with example*.

ScholarHat. <https://www.scholarhat.com/tutorial/datastructures/one-dimensional-array>

TutorialsPoint. (n.d.). *Data structures and algorithms (DSA) tutorial*.

https://www.tutorialspoint.com/data_structures_algorithms/index.htm